SAMPLE

Earth's Sky: Transits of Venus and Mercury

Planets Crossing the Sun

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Introduction

Twice in a hundred and twenty years, give or take, the planet Venus crosses the face of the Sun as seen from the Earth: a small black disc, a fraction of the Sun's own width, sliding across the blazing circle for some six hours, an event so rare that most human generations see it never, and so precise in its geometry that the species spent three centuries using it to measure the solar system. Mercury makes the same crossing far more often, thirteen or fourteen times a century, a smaller dot on a quicker pass; and these two events, the transits of the inner planets, are the subject of this book: the rarest predictable sight in the sky, the astronomical unit's first honest measurement, and the ancestor of the method that has since found the planets of other stars.

The story is one of the best in the history of science. It begins with Kepler, whose tables first predicted the crossings, and with young Jeremiah Horrocks, the English clergyman who corrected the great man's arithmetic and caught the Venus transit of 1639 from a Lancashire window, the first person in recorded history to see one. It runs through Edmond Halley, who saw that the transit's timing, measured from different places on Earth, would yield the Sun's distance, and called on future astronomers to mount the expeditions he would not live to lead; through the great international campaigns of 1761 and 1769, the first worldwide scientific enterprises, Captain Cook's voyage to Tahiti among them, and the century's best measurement of the astronomical unit; through the photographic expeditions of 1874 and 1882, the method's last professional engagements; and to the final pair of 2004 and 2012, watched by hundreds of millions, the last Venus transits anyone now alive will see.

The book follows the whole arc. The chapters describe what a transit is and why it happens; the orbital geometry that makes the events so rare, the crossing of the nodes and the pairing of the Venus years; the history of prediction and observation from Kepler to the space age; the measurement's method and the expeditions it drove, with their triumphs, their tragedies, and the black drop that fooled them all; the discoveries made along the way, including Venus's atmosphere, found in the transit's own light; and the modern afterlife, the transit method become the exoplanet hunter's chief tool, the small black dot of 1639 grown into the key to thousands of worlds.

A remark on the book's coverage and its neighbors, before the crossing's story starts. The account keeps to the transits' own ground: the local crossings of the inner planets, their history, their geometry, and their method's descendants, with the exoplanet transit-survey described as the inheritance rather than as a rival book's whole subject, since the wider exoplanet story fills its own volume in this series. The observational instruction is kept safe and simple, the projection and the certified filter the only methods taught, the Sun's danger stated once and honored throughout. The numbers quoted, the transit dates and the cycle's intervals, the disc diameters and the measured unit, are the record's own established values, and the stories of the observers, Horrocks's Sunday and Cook's voyage and the cloud-watched decade, are the biography the subject's own literature has kept. The reader completing the chapters holds the transits whole: the geometry, the history, the method, and the calendar's honest arithmetic, the rarest sight's complete account in plain words.

One further preparatory remark, the account's own arrangement. The sixteen sections run in the series' customary sequence: the discovery and the understanding open, the characteristics and the science follow, the observation, the features, and the contexts fill the middle, the modern understanding, the research, and the discoveries come next, and the difficulties, the forward looks, the lighter lore, and the close end the volume; and each section reads on its own for whoever opens at the page of interest. The technical vocabulary is explained where it first appears and reused thereafter, the story's terms built as the account proceeds; the dates and the numbers are the record's settled values, quoted plainly; and the meanings this introduction has promised wait in the closing chapter rather than being asserted through the evidence. The reader who wants the whole story can read straight through, and the reader with one question, why the transits pair off, or what the black drop was, will find its answer in its section's paragraphs: a reference work as well as a narrative, the rarest sky's account arranged for the cover-to-cover reading and the single lookup both.

The reader should carry from the start the transit's double character: it is at once the gentlest and the grandest of celestial events, a speck crossing a fire, requiring nothing but the right day, a safe filter, and patience to observe, and yet carrying the scale of the cosmos, the distance of the Sun, and now the census of the galaxy's planets in its modest geometry. The story of the transits is the story of how the species learned to use the rarest alignments as its finest instruments, and it is told here whole, from Horrocks's window to the telescope fleets of the exoplanet era.

Chapter 1: The Discovery of Transits of Venus and Mercury

The transits, in the sense of the events themselves, are as old as the solar system; the discovery, in the sense of the knowing, has a date and a name, and it belongs to the generation that first trusted the sky's arithmetic enough to wait for it. The possibility came with the heliocentric order: once the planets were understood as worlds circling the Sun, inside the Earth's own orbit, the crossing of the inner planets across the solar face was a geometric consequence, waiting only for the tables good enough to say when. Those tables came with Kepler, whose laws and whose Rudolphine calculations of the planetary positions put the planets' paths on the ecliptic with a precision no predecessor matched; and in the calculations for 1631 the transits of both Mercury and Venus appeared, the first predicted crossings in the history of the subject, Kepler warning his readers to watch the Sun's face on the appointed November morning.

Mercury went first. Pierre Gassendi, the Parisian observer, watched the predicted day of 1631 with a projected solar image, the safe method this book will teach again in a later chapter, and saw the small planet enter the Sun's disc, the first transit ever observed, a modest black point whose timing he recorded, the prediction's verification. Venus was to cross a month later, but the December date fell after the Paris winter's dawn problem: the transit, as it happened, was not visible from Europe's longitudes anyway, and the world missed it, Kepler's own tables erring in the very year they were proved right on Mercury.

The Venus transit's discovery, four years later, is the subject's most famous story. Kepler's tables, for all their excellence, put the Venus crossings of the era off by days; and the error was caught by Jeremiah Horrocks, a young astronomer in his early twenties, working from a Lancashire curacy, who reworked the planet's orbit from the observations and found the tables putting a transit where none was listed: the crossing would come, on his corrected arithmetic, on the fourth of December, 1639, a Sunday, in the afternoon. Horrocks set up his observing, a small telescope and the projected image, the church duties interleaved with the watch; and near mid-afternoon the small black disc of Venus entered the Sun's face, the first Venus transit ever seen, observed also by his correspondent William Crabtree in Manchester, who was so startled at the sight that he nearly failed to record it. The observation's whole tone, in Horrocks's letters, is the modern one: the prediction made from the physics, the event caught at the appointed hour, the sky's machinery trusted and vindicated, the birth of the transit's history in one English afternoon.

A further thread of the discovery's background deserves its own paragraph, the astronomical culture that made the prediction possible. The transit's prediction required the heliocentric order, the elliptical orbits, and the tables' precision, and each of these was a hard-won victory of the preceding century: Copernicus's rearrangement, the Earth made a planet among planets, the inner crossings made geometrically possible at a stroke; Tycho's observations, the sky's sharpest pre-telescopic record, the data the laws were fitted to; Kepler's own struggle, the orbit-shapes found in the Mars data's stubbornness, the ellipse wrested from the circles of two thousand years; and the logarithms, the calculation's own tool, invented for the tables' sake, the Rudolphine numbers computed by the new arithmetic. The thread's meaning is the prediction's own cost: the 1631 transit was foretold by a whole century's work, the small dot on the Sun the visible proof of an invisible revolution, and the discovery chapter's story, told whole, begins not with Kepler's printing but with the century's patient rebuilding of the sky's order that made his numbers possible at all.

The discovery chapter earns one more entry, the telescope's own arrival, the instrument that made the transit an observation. The crossing of 1631 and 1639 fell in the instrument's first decades: the spyglass's birth in the Dutch trade, Galileo's skies, the Sun's own face first examined in the same era, the projected-image method invented for the solar observation's very safety, the whole toolkit of the transit's watch created in the years between the first predictions and the first observations; and Gassendi's and Horrocks's equipment, the modest refractors and the camera-obscura projections, were the newest instruments of their day, the small dot's first witnesses using the age's newest eyes. The entry's meaning is the timing's own rightness: the transit's discovery required the tables and the telescope together, the arithmetic and the optics arriving in the same generation, and the 1630s, holding both, made the sky's rarest sight an observed fact at the first opportunity, the subject's birth in the instruments' golden dawn.

The discovery's meaning, held plainly, is the transit's own standing in the science's progress: from 1631 and 1639 onward, the crossings were known events, computable and verifiable, and the next step of the story, the measurement, could begin to be planned. Halley, a generation later, saw the method in the transit's geometry; the expeditions followed the method; and the chapters that follow trace the whole line, from Horrocks's afternoon to the worldwide campaigns and the modern afterlife, the discovery's small start growing into the subject this book records.